



Mini-BEHAVIOR-Gran: Revealing U-Shaped Effects of Instruction Granularity on Language-Guided Embodied Agents

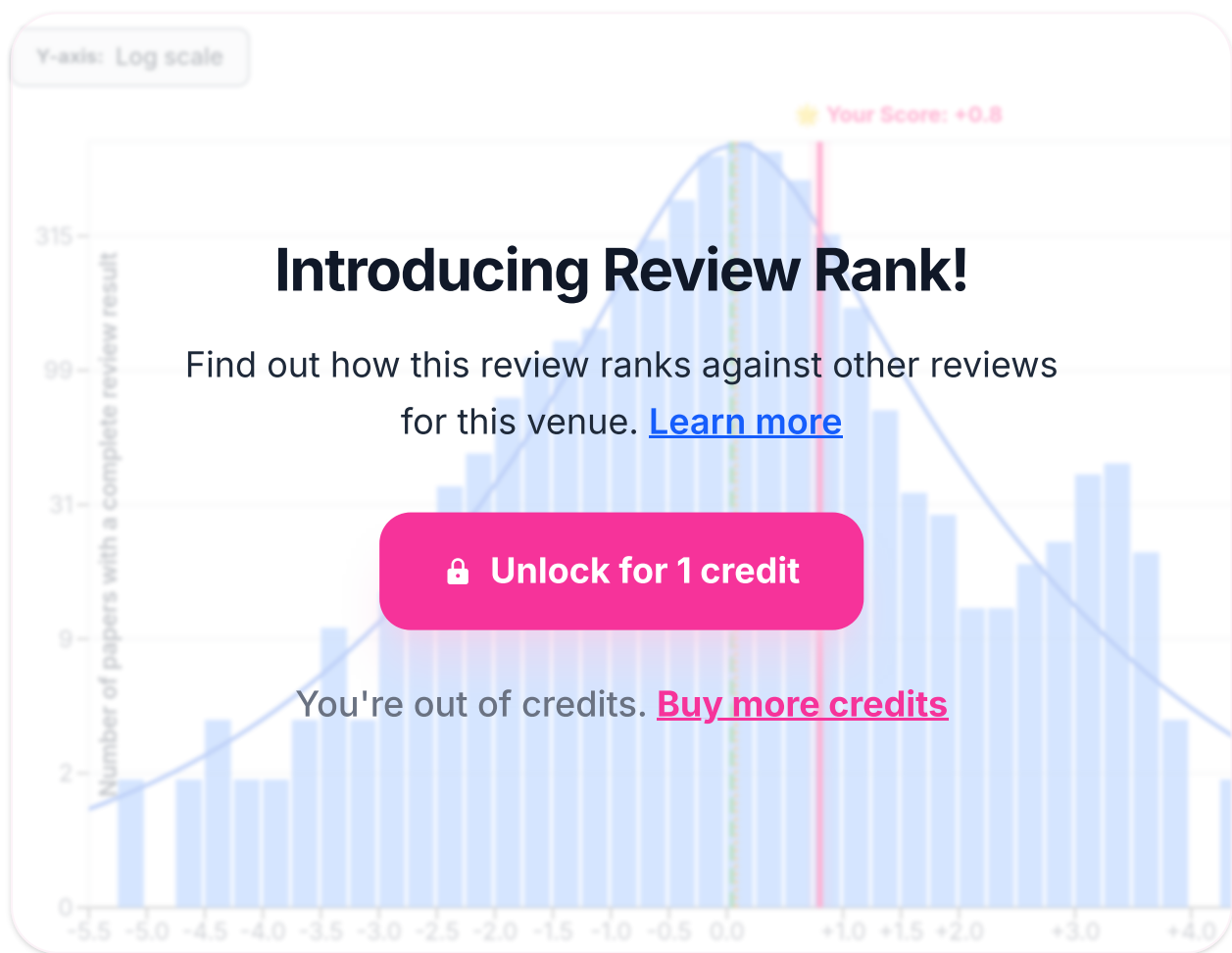
Anonymized Authors

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Reviewed as if submitted to CONFERENCE ACL 2026 - main track ⓘ



Desk Rejection Assessment:

Paper Length

Pass .

Topic Compatibility

Pass . The paper investigates the impact of language instruction granularity on embodied agents, a topic that directly aligns with the ACL area of "Multimodality and Language Grounding to Vision, Robotics and Beyond".

Minimum Quality

Pass . The paper is structurally complete, including an abstract, introduction, related work, methodology, experiments, results, and conclusion. The scientific methodology appears sound for a controlled study, and the writing is clear and meets academic standards.

Prompt Injection and Hidden Manipulation Detection

Pass . No evidence of prompt injection or hidden manipulation was found in the paper content.

Expected Review Outcome:

Paper Summary

This paper introduces MINI-BEHAVIOR-GRAN, a new benchmark for studying the effect of instruction granularity on language-guided embodied agents. The benchmark extends MINI-BEHAVIOR by providing multiple instruction variants for each task, ranging from high-level goals to fine-grained steps. The authors propose and validate "width," a metric from automated planning, as a consistent cross-task measure of granularity. Using this benchmark, the paper reveals a U-shaped relationship between instruction granularity and agent performance: agents perform well with very fine-grained and very coarse instructions, but struggle with intermediate levels. The authors attribute the performance rebound at the coarse end to agents learning "shallow grounding" policies that rely more on visual cues than on language.

Summary Of Strengths

The paper presents a well-executed and focused study on an important but under-explored variable in embodied AI: instruction granularity. Its primary strengths are:

1. **Novel and Controlled Benchmark:** The main contribution, MINI-BEHAVIOR-GRAN, is a valuable resource. By systematically generating instruction



variants from a shared symbolic layer, it enables controlled experiments that are difficult to conduct with existing datasets. The dynamic instruction generation, which updates guidance based on the agent's state, is a notable improvement over static instructions.

2. **Principled Granularity Metric:** The paper makes a strong case for using "width" as a cross-task metric for granularity. The empirical validation in **Figure 4**, showing that width has a more consistent negative correlation with success rate than surface-level metrics like token or entity count, is convincing. This provides the community with a more principled way to quantify an instruction's latent planning burden.
3. **Interesting and Non-Obvious Finding:** The discovery of a U-shaped performance curve (**Figure 5**) is a significant finding. It challenges the intuitive assumption that performance should monotonically decrease as instructions become coarser. This result is thought-provoking and opens up new questions about the trade-offs in agent learning.
4. **Insightful Analysis of the U-Shaped Effect:** The paper doesn't just report the U-shaped curve; it provides a compelling, evidence-backed explanation. The hypothesis of "shallow grounding" is well-supported by a series of targeted experiments, including the instruction predictability analysis (**Figure 6**) and the language ablation studies (**Figure 7**). This analysis, which connects the performance rebound to a shift towards vision-dominant policies, is a major strength.
5. **Thorough Experimental Design:** The study is methodologically sound. The authors evaluate three distinct VLA action-decoding archetypes, demonstrating that their findings are not specific to one architecture. The two-stage experimental protocol and the various mixed-granularity training settings (**Table 5**) provide a comprehensive view of the learning dynamics.

Summary Of Weaknesses

Despite its strengths, the paper has several weaknesses that temper its impact and raise questions about the generalizability of its conclusions.

1. **Over-reliance on a Symbolic, Grid-World Environment:** The most significant weakness is the choice of the MINI-BEHAVIOR environment. While the authors justify this choice for control and isolation (Appendix C), the grid-world nature and reliance on a predefined symbolic feature space (Φ) severely limit the external validity of the findings. The entire framework for generating instructions and computing the "width" metric is contingent on this symbolic layer, which is unavailable in more realistic, pixel-based environments. It is unclear if the "width" metric, or the U-shaped effect itself, would manifest in environments with continuous state/action spaces and instructions grounded in natural, unstructured language.



2. **Artificiality of Instruction Generation:** The instructions are generated from rule-based templates. While this ensures control, it results in language that may not reflect the diversity and ambiguity of human-generated instructions. The paper claims "moderate linguistic complexity" (Section 2.4), but this is a stretch. The template-based generation process likely creates a strong correlation between the symbolic rules and the surface text, which might make the "shallow grounding" problem more pronounced than it would be with natural human language.
3. **Ambiguity in the "Width" Metric's Practicality:** The paper successfully argues for "width" as an analytical tool within its controlled benchmark. However, its definition relies on having access to the ground-truth symbolic features and an optimal plan to compute the number of concurrently tracked features (**Table 3**). The authors concede this limitation in Section 7, stating it's "not as a deployable measure for real-world settings." This is a critical point that should be emphasized earlier. Without a clear path to even *approximating* width in real-world scenarios, its utility beyond this specific study is questionable.
4. **Potential Confounding in the Coarse Granularity Grouping:** The paper partitions width into Fine ($w=0$), Medium ($w=1$), and Coarse ($w \geq 2$) for its main analysis. However, **Table 5** reveals that the "Coarse" group is dominated by $w=2$ instances, which appear in 19 tasks, while higher-width instances ($w=3$ in 10 tasks, $w \geq 4$ in 3 tasks) are much rarer and show sharply declining performance. Aggregating these into a single "Coarse" category might present a misleadingly optimistic picture of performance rebound. The rebound seems to be a $w=2$ phenomenon, followed by a collapse. This nuance is somewhat lost in the main narrative built around **Figure 5**.
5. **Unconvincing Attention Analysis:** The layer-wise attention analysis in **Table 4** is presented as supporting evidence for the shallow grounding hypothesis, but it feels weak and post-hoc. The binomial test on attention trends is an indirect probe, and the results are inconsistent across architectures (e.g., "Parallel models lack consistent trends"). The claim that a decrease in language attention implies an increase in visual reliance is not straightforward, as the authors themselves note that action tokens can attend to other tokens. This analysis does not provide strong mechanistic evidence.
6. **Limited Scope of Granularity:** The paper's definition of granularity is exclusively focused on subgoal decomposition. As acknowledged in the limitations, other important dimensions like vagueness ("near the table"), linguistic abstraction ("repeat until clean"), and ambiguity are not considered. While this focus allows for a clean study, it also means the paper is investigating only one facet of a multi-dimensional problem.
7. **Clarity of the Dynamic Instructor Module:** The pipeline in **Figure 3** is complex. The "Dynamic Rule Selection" part of the instructor module is not

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fully explained. It states that it "Randomly select one from available rules r_t in $\mathcal{R}_a$ ". If multiple rules are applicable, how is one chosen? Is it random? If so, this introduces stochasticity into the instruction stream that isn't discussed. If there's a deterministic policy, it should be described. This ambiguity affects the reproducibility and interpretation of the agent's learning process.

Potentially Missing Related Work

The related work section is reasonable but could be strengthened by discussing recent works on generating or leveraging diverse instructions for embodied agents. These papers are relevant to the core theme of how instruction properties affect agent learning.

- **Wang, Z., Li, J., Hong, Y.** (2024): *Bootstrapping Language-Guided Navigation Learning with Self-Refining Data Flywheel*. This work focuses on generating high-quality instruction-trajectory pairs and could be discussed in Section 5 in the context of data generation for training robust agents.
- **Gopinathan, M., Masek, M., Abu-Khalaf, J.** (2024): *Spatially-Aware Speaker for Vision-and-Language Navigation Instruction Generation*. This paper develops an instruction generator that uses environmental knowledge, which is relevant to the discussion of how instruction content (and its granularity) is shaped and how it impacts navigation.
- **Li, D., Zhao, C., Yang, S.** (2024): *Learning Instruction-Guided Manipulation Affordance via Large Models for Embodied Robotic Tasks*. This explores how language guides manipulation, providing a point of comparison for how instruction detail affects performance in manipulation vs. navigation tasks. It should be cited in Section 5.
- **Goel, D., Singh, K. P., Choi, J.** (2022): *Language Guided Meta-Control for Embodied Instruction Following*. Proposes a meta-controller for better language grounding, which is highly relevant to the paper's theme of overcoming grounding challenges. It could be discussed in Section 5 as an alternative approach to improving robustness.
- **Sarch, G., Wu, Y., Tarr, M.** (2023): *Open-Ended Instructable Embodied Agents with Memory-Augmented Large Language Models*. This work on agents that adapt to user instructions is relevant to the discussion of how agents handle diverse instruction styles, a key motivation for this paper.
- **Dorbala, V. S., Chowdhury, S., Manocha, D.** (2024): *Can LLM's Generate Human-Like Wayfinding Instructions? Towards Platform-Agnostic Embodied Instruction Synthesis*. This investigates using LLMs to generate instructions, which contrasts with the template-based approach in the submission and is relevant for the discussion on instruction sources in Section 2.

Comments Suggestions And Typos

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1. **Clarify the "Coarse" Grouping:** I suggest the authors be more upfront about the composition of the "Coarse" granularity group. In Section 4.3, when introducing the U-shaped effect, it would be more transparent to immediately reference the per-width breakdown in **Table 5** and acknowledge that the rebound is primarily driven by $w=2$ before performance collapses at higher widths. This would paint a more nuanced and accurate picture.
2. **Strengthen the Link to Practicality:** While the "width" metric is currently analytical, could the authors speculate on how it might be approximated in the future? For example, could an LLM be used to parse an unstructured instruction into a pseudo-symbolic plan, from which an approximate width could be estimated? Discussing such future directions would help bridge the gap between this controlled study and real-world applications.
3. **Dynamic Instructor Logic:** Please clarify the logic for selecting the active rule r_t when multiple rules in $\mathcal{R}_a$ are applicable (Section 2.3 and **Figure 3**). Is it random, or is there a heuristic? This is important for reproducibility and for understanding the data distribution the agent is trained on.
4. **Typos/Minor Issues:**
 - In Table 1, the feature pr is listed twice with different descriptions ("distance to the nearest rag" and "distance to the sink"). This should be corrected (e.g., ps for sink).
 - In Section 2.2, the text refers to rules r_{1a}, r_{1b}, r_{1c} in **Figure 2**, but the figure itself shows these being merged into a coarser rule r_1 . The text says they "form a coherent subplan r_1 that abstracts into a single coarse rule in $\mathcal{R}^{(3)}$ ", which is a bit confusing. It might be clearer to say that the atomic rules in $\mathcal{R}^{(2)}$ are merged to form the macro-rule in $\mathcal{R}^{(3)}$.
 - The citation style is inconsistent in some places (e.g., (Chevalier-Boisvert et al., 2023) vs. (Gao et al., 2025) with an asterisk). Please ensure consistency.

A response clarifying these points, especially the first three, would strengthen the paper and could improve my assessment. Specifically, a more nuanced discussion of the coarse-end rebound and a clearer path toward making "width" a more practical concept would be very valuable.

Confidence

4 Quite sure after careful checking: I tried to check the important points carefully. It's unlikely, though conceivable, that I missed something that should affect my ratings.

Soundness

3.5 Between Acceptable and Strong: This study provides sufficient support for its main claims. Some minor points may need extra support or details. The core methodology is sound for a controlled study, but the reliance on a symbolic grid-world limits the generality of the claims.

Excitement

3.5 Between interesting and exciting: I would mention this paper to others and/or make an effort to attend its presentation in a conference. The U-shaped finding and the "shallow grounding" explanation are genuinely interesting and thought-provoking.

Overall Assessment

[See how you rank ↑](#)

4 Conference: This paper could be accepted to an ACL conference. It demonstrates sufficient novelty, soundness, and impact. The paper introduces a useful benchmark, a principled metric, and a non-trivial finding about agent behavior, all supported by a solid, well-designed set of experiments. While the reliance on a simplified environment is a notable limitation, the insights gained from this controlled study are valuable and likely to stimulate further research into instruction granularity. The strengths in its focused investigation and clear findings outweigh the weakness of limited external validity, making it a solid contribution to the main conference.

Best Paper Justification

N/A

Limitations And Societal Impact

The authors have provided a dedicated "Limitations" section (Section 7) that is commendably thorough and transparent. They correctly identify the key limitations of their work: the use of a discrete action space, the narrow definition of granularity focusing on subgoal decomposition, and the impracticality of computing the "width" metric in real-world settings. This is a well-written section that appropriately contextualizes the paper's contributions and sets clear directions for future work.

The paper does not discuss societal impact, which is acceptable given its foundational research nature. The work does not pose immediate negative societal risks. Potential positive impacts are indirect, through advancing the understanding needed to build more robust and reliable language-guided robots that can interact more naturally with humans.

Ethical concerns

None. The work is conducted in a simulated environment and does not involve human data, privacy concerns, or other direct ethical issues. The authors have included a statement on LLM usage in the appendix.

Needs Ethics Review

No.

Reproducibility

4 Mostly reproducible: They could mostly reproduce the results, but there may be some variation because of sample variance or minor variations in their interpretation of the protocol or method. The authors plan to release the benchmark, which is excellent. The methodology is described in detail, though a minor ambiguity exists in the dynamic instructor module's selection logic. Given the code and benchmark, reproduction should be feasible.

Datasets

4 Useful: I would recommend the new datasets to other researchers or developers for their ongoing work. The MINI-BEHAVIOR-GRAN benchmark will be a valuable resource for researchers specifically interested in controlled studies of language grounding, instruction following, and agent robustness to linguistic variation.

Software

1 No usable software released. The paper does not mention releasing the code for the VLA models or the experimental pipeline, only the dataset. Releasing the full codebase would increase the value and reproducibility.

Disclaimer: Only the main paper is reviewed; anonymity and formatting are not checked. The desk-rejection assessment evaluates content completeness, correctness, novelty, clarity, and scientific quality in line with common ACL review guidelines.

Reviewed on March 15, 2026  Gemini-2.5-Pro

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